

Lecture 1: Introduction to Web Development & HTML5 Basics

Duration: 3 Hours

Target Audience: Beginners

1. What is Web Development?

Web development is the process of building websites and applications for the internet. Think of building a house:

- **HTML (The Structure):** The bricks, walls, and layout of the house.
- **CSS (The Style):** The paint, decorations, and interior design.
- **JavaScript (The Functionality):** The electricity, plumbing, and moving parts (like doors opening).

Today, we are starting with the foundation: HTML.

2. What is HTML?

HTML stands for **HyperText Markup Language**.

- **HyperText:** Links that connect web pages together.
- **Markup Language:** A way to use "tags" to tell the web browser how to display text, images, and other content.

3. Hello World & Basic Website Structure

Every HTML document has a specific structure. Let's write our first "Hello World" webpage!

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>My First Webpage</title>
</head>
<body>
  <h1>Hello World!</h1>
</body>
</html>
```

Understanding the Viewport Tag

Notice the `<meta name="viewport" content="width=device-width, initial-scale=1.0">` tag in the head. This tells the browser how to control the page's dimensions and scaling. It is absolutely necessary to make your website "responsive" so that it looks good on both big computer screens and small mobile phones.

4. Basic Tags, Headings, and Paragraphs

Heading Tags

HTML has 6 levels of headings, from `<h1>` (most important/largest) to `<h6>` (least important/smallest).

```
<h1>Main Title of the Page</h1>
<h2>Subheading</h2>
<h3>Smaller Section Heading</h3>
```

Paragraph Tag

The `<p>` tag is used for normal blocks of text.

```
<p>This is a normal paragraph of text on our website. It is used
for standard reading content.</p>
```

5. Attributes and Comments

HTML Attributes

Attributes provide extra information about HTML elements. They always go inside the opening tag and come in name/value pairs like `name="value"`.

```
<!-- Example of an href attribute in a link -->
<a href="https://www.google.com">Click here to visit Google</a>

<!-- Example of src and alt attributes in an image -->

```

HTML Comments

Comments are notes you write in the code that the browser ignores. They are strictly for developers to read.

```
<!-- This is a comment. The user won't see this on the webpage! Use  
it to leave notes for yourself. -->
```

6. Block vs. Inline Elements

Elements in HTML generally fall into two display categories:

- **Block-level elements:** These always start on a new line and take up the full width available from left to right. Examples: <h1>, <p>, <div>.
- **Inline elements:** These do not start on a new line and only take up as much horizontal space as their content needs. Examples: , <a>, .

7. Lists in HTML

We use lists to group related items together. Every item inside a list is wrapped in an (List Item) tag.

Unordered Lists ()

Used when the order doesn't matter (displays as bullet points).

```
<ul>  
  <li>Apples</li>  
  <li>Bananas</li>  
  <li>Oranges</li>  
</ul>
```

Ordered Lists ()

Used when the order matters (displays as numbers by default). You can use attributes like type and start to change how they count.

```
<ol type="A" start="3">  
  <!-- This will start numbering from 'C' -->  
  <li>First item (Will show as C)</li>  
  <li>Second item (Will show as D)</li>  
</ol>
```

8. Tables in HTML

Tables are used to display data in structured rows and columns.

- `<table>`: Wraps the entire table.
- `<thead>`: Groups the header content at the top.
- `<tbody>`: Groups the main body content of the table.
- `<tr>`: Table Row (creates a new horizontal row).
- `<th>`: Table Header (makes text bold and centered).
- `<td>`: Table Data (creates a standard cell).

Rowspan and Colspan

Sometimes you need a cell to stretch across multiple rows or columns.

```
<table border="1">
  <tr>
    <th colspan="2">This Heading Stretches Across 2
Columns</th>
  </tr>
  <tr>
    <td rowspan="2">This cell stretches down 2 rows</td>
    <td>Normal Cell 1</td>
  </tr>
  <tr>
    <td>Normal Cell 2</td>
  </tr>
</table>
```

9. HTML Forms

Forms are used to collect data from the user (like login forms or contact forms).

- `<form>`: The container for all input fields.
- `<label>`: A text label describing an input.
- `<input>`: The actual field where users type. The type attribute changes its behavior.
- `<button>`: Used to submit the form.

```
<form action="/submit-data">
  <label for="username">Username:</label>
  <input type="text" id="username" name="username"
placeholder="Enter name">

  <label for="password">Password:</label>
```

```
<input type="password" id="password" name="password">
```

```
<button type="submit">Log In</button>  
</form>
```

10. HTML Media Tags & Embeds

HTML5 makes it incredibly easy to add audio, video, and content from other websites.

Audio and Video

Use the `<audio>` and `<video>` tags with the `controls` attribute to give users play/pause functionality.

```
<!-- Adding an Audio track -->  
<audio controls>  
  <source src="music.mp3" type="audio/mpeg">  
  Your browser does not support the audio element.  
</audio>
```

```
<!-- Adding a Video -->  
<video width="600" controls>  
  <source src="presentation.mp4" type="video/mp4">  
  Your browser does not support the video tag.  
</video>
```

Iframes (Embedding External Content)

An `<iframe>` is used to embed another document within the current HTML document. This is how you embed Google Maps or YouTube videos.

```
<!-- Embedding a YouTube Video -->  
<iframe width="560" height="315"  
  src="https://www.youtube.com/embed/dQw4w9WgXcQ"  
  title="YouTube video player" frameborder="0"  
  allowfullscreen>  
</iframe>
```

11. Assignments

Assignment 1: Create a Basic Webpage

Task: Create a basic HTML page. It must include:

- A main heading (<h1>) and a subheading (<h2>).
- At least two paragraphs (<p>) of text about your favorite hobby.
- An image related to your hobby (using the tag with src and alt attributes).
- A link (<a>) to a website where you can learn more about the hobby.

Assignment 2: Class Schedule Table (Vertical Break)

Task: Create a class schedule for Monday to Friday using HTML tables. You must include a "LUNCH BREAK" that stretches vertically across all 5 days using the rowspan attribute.

```
<table border="1" cellpadding="10" cellspacing="0">
  <thead>
    <tr>
      <th>Time</th>
      <th>9:00 - 11:00 AM</th>
      <th>11:00 - 1:00 PM</th>
      <th>1:00 - 2:00 PM</th>
      <th>2:00 - 4:00 PM</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Monday</td>
      <td>Math</td>
      <td>English</td>
      <td rowspan="5" style="text-align: center; font-weight:
bold;">LUNCH<br>BREAK</td>
      <td>Computer Science</td>
    </tr>
    <tr>
      <td>Tuesday</td>
      <td>Physics</td>
      <td>History</td>
      <!-- Lunch cell omitted here due to rowspan -->
      <td>Lab</td>
    </tr>
    <tr>
      <td>Wednesday</td>
      <td>Math</td>
      <td>English</td>
      <td>Computer Science</td>
    </tr>
    <tr>
      <td>Thursday</td>
      <td>Chemistry</td>
```

```
        <td>Biology</td>
        <td>Lab</td>
    </tr>
    <tr>
        <td>Friday</td>
        <td>Physics</td>
        <td>History</td>
        <td>Physical Ed</td>
    </tr>
</tbody>
</table>
```

Assignment 3: Multimedia Webpage

Task: Build a single-page profile or fan page for your favorite movie, band, or video game.

Requirements:

1. Use proper structural tags (headings and paragraphs).
2. Embed a YouTube video related to the topic using an <iframe>.
3. Add an <audio> tag that plays a related sound clip or song (you can use a dummy file name if you don't have a real one, but the structure must be correct).
4. Include a basic <form> at the bottom allowing users to "Subscribe for Updates" (requires a text input for name, email input, and a submit button).